Exploratory Data Analysis Report

E-Commerce Sales Dataset

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1. Project Overview

This report presents the findings of an Exploratory Data Analysis (EDA) conducted on an e-commerce sales dataset. The primary objective was to understand the structure of the data, detect and handle missing values, compute descriptive statistics, and visualise sales patterns across products and price distributions. The analysis was implemented in Python using the pandas, matplotlib, and seaborn libraries.

2. Dataset Description

The raw dataset contained **1,200 records** and **14 columns**, spanning orders placed between January 2023 and June 2025. Each record represents a single customer order and includes the following attributes:

Column	Description
OrderID	Unique identifier for each order
Date	Order date and timestamp
CustomerID	Unique customer identifier
Product	Product category (Chair, Desk, Laptop, Monitor, Phone, Printer, Tablet)
Quantity	Number of units ordered (1–5)
UnitPrice	Price per unit (USD)
ShippingAddress	Delivery address
PaymentMethod	Method used for payment
OrderStatus	Current status of the order
TrackingNumber	Shipment tracking code
ItemsInCart	Total items in cart at checkout (1–10)
CouponCode	Discount coupon applied (if any)
ReferralSource	Marketing channel (Email, Facebook, Instagram, Referral)
TotalPrice	Final order value in USD

3. Data Cleaning

Prior to analysis, the dataset was inspected for missing values. The audit revealed that only the **CouponCode** column contained missing entries (309 nulls out of 1,200 records, approximately 25.75%). All other columns were complete with zero missing values. The missing coupon codes were imputed with the placeholder value

'No Coupon', preserving all rows for downstream analysis. After cleaning, the dataset was fully complete with no remaining nulls.

Metric	Before Cleaning	After Cleaning
Total Records	1,200	1,200
Columns with Missing Values	1 (CouponCode)	0
Missing Values (CouponCode)	309	0
Rows Dropped	—	0

4. Descriptive Statistics

The table below summarises the key statistical measures for the four numerical variables in the cleaned dataset:

Statistic	Quantity	Unit Price (USD)	Items in Cart	Total Price (USD)
Count	1,200	1,200	1,200	1,200
Mean	2.95	356.41	5.49	1,053.97
Median	3.00	364.21	5.00	823.62
Std Dev	1.41	197.18	2.28	819.86
Min	1.00	11.39	1.00	11.39
25th Pct	2.00	186.06	4.00	410.52
75th Pct	4.00	521.57	7.00	1,578.48
Max	5.00	699.93	10.00	3,456.40

Key observations: The average order value of USD 1,053.97 is notably higher than the median (USD 823.62), indicating a right-skewed distribution driven by high-value orders. The standard deviation of USD 819.86 reflects considerable spread in total price, consistent with the wide product mix. Quantity per order is relatively uniform (mean ≈ 3), while items in cart average approximately 5 per session.

5. Data Visualisations

5.1 Total Sales by Product

The bar chart below shows the aggregated total revenue (sum of TotalPrice) for each of the seven product categories. Sales are broadly comparable across all products, ranging from approximately USD 152,000 (Phone) to USD 196,000 (Chair and Printer), suggesting a well-balanced product portfolio with no single dominant category.

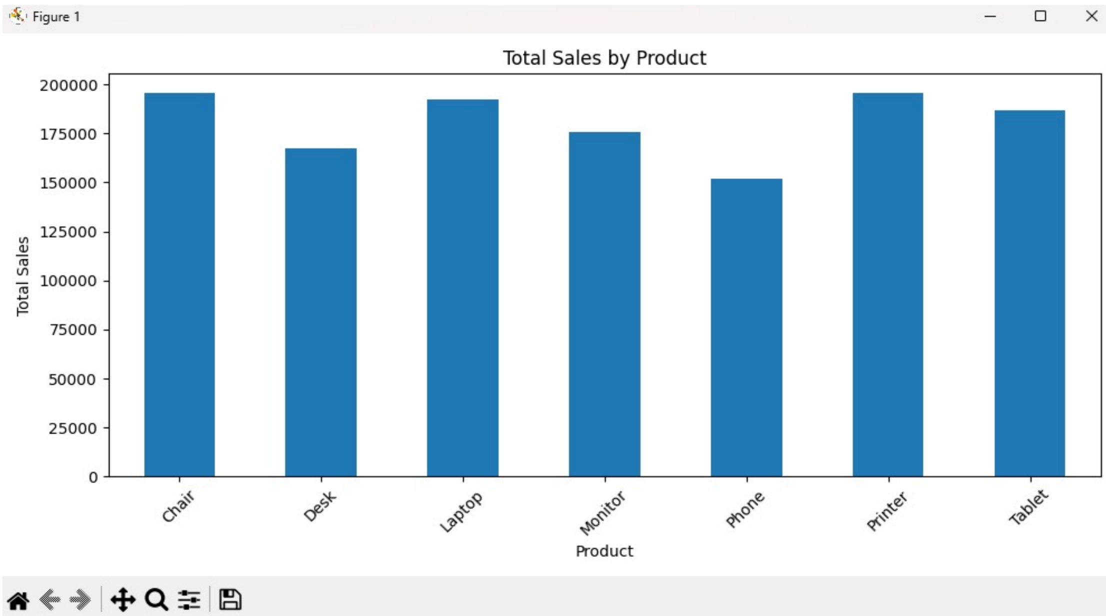

Figure 1 – Total Sales Revenue by Product Category

Product	Approx. Total Sales (USD)	Relative Rank
Chair	~195,000	1st
Printer	~195,000	2nd
Laptop	~192,000	3rd
Tablet	~185,000	4th
Monitor	~175,000	5th
Desk	~167,000	6th
Phone	~152,000	7th

5.2 Distribution of Total Price

The histogram with KDE overlay illustrates the frequency distribution of TotalPrice across all 1,200 orders. The distribution is clearly right-skewed (positive skew), with the majority of orders concentrated in the USD 0–700 range and a long tail extending to USD 3,456. The KDE curve confirms the absence of a normal distribution, which is typical for transaction-level revenue data.

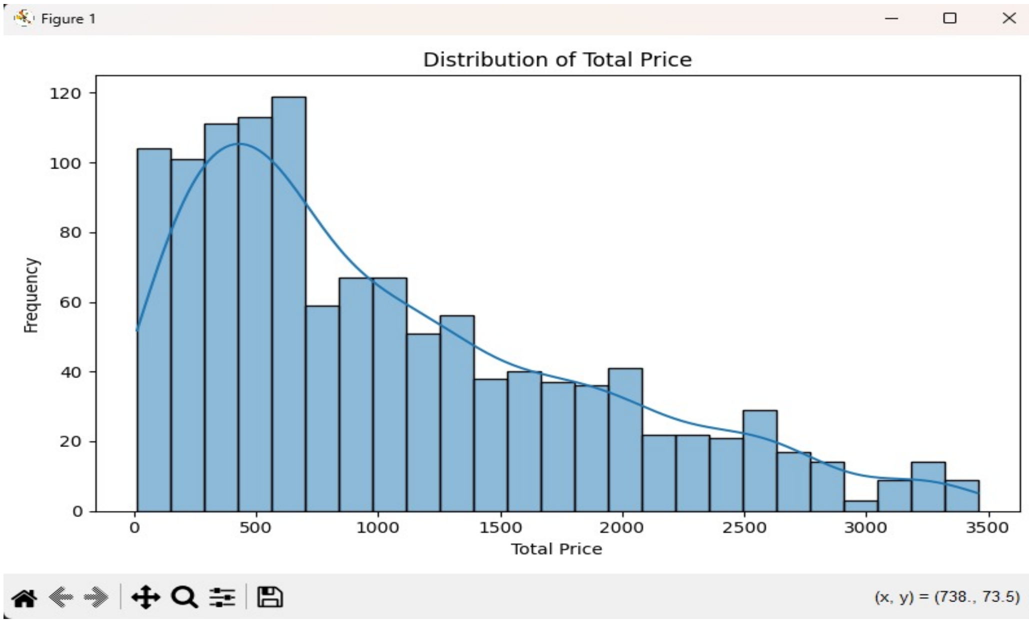

Figure 2 – Distribution of Total Price (Histogram + KDE)

6. Key Findings & Insights

Data Quality: The dataset was 97.4% complete. Only CouponCode had missing values (25.75%), cleanly resolved by imputation. No records were dropped, retaining the full 1,200-row dataset.

Revenue Distribution: Total Price is right-skewed with a mean of USD 1,054 vs. median of USD 824. High-value outlier orders (up to USD 3,456) pull the mean upward, suggesting a small segment of high-spend customers.

Product Performance: Chair and Printer led total revenue (~USD 195K each), while Phone had the lowest revenue (~USD 152K). The narrow spread (~USD 43K) across all 7 products indicates a balanced catalogue.

Order Behaviour: The average quantity per order is ~3 units with items in cart averaging ~5, suggesting customers often browse more products than they ultimately purchase.

Price Range: Unit prices span from USD 11.39 to USD 699.93, offering a wide range that caters to different customer budget segments.

7. Tools & Methodology

Tool / Library	Version	Purpose
Python	3.x	Core programming language
pandas	latest	Data loading, cleaning, and aggregation
matplotlib	latest	Bar chart generation
seaborn	latest	Histogram with KDE overlay
VS Code	latest	Development environment
reportlab	latest	PDF report generation

8. Conclusion

This EDA successfully characterised the e-commerce sales dataset across all key dimensions: data quality, univariate statistics, and product-level revenue performance. The data is largely clean and ready for further modelling. Future analysis could explore temporal sales trends, customer segmentation by referral source, the impact of coupon usage on order value, and predictive modelling of high-value customers.